



C & C++ Programming

CERTIFICATION COURSE

**Master C & C++ Programming Skills and
Elevate Your Career to the Next Level!**

C & C++ Programming

The C and C++ Programming Course, offered by W3 Technologies, will equip you with the skills to write efficient, high-performance applications using C and C++. You'll learn to break down complex programming challenges into manageable components, focusing on C and C++ syntax, memory management, and best practices for clear, maintainable code. This course emphasizes algorithm optimization, data structures, and modular design, providing a solid foundation in C and C++ development.

Why W3 Technologies

Training at W3 Technologies equips you to excel in today's competitive world. From the very beginning of your journey with us, you'll experience continuous growth through learning. We provide support throughout the course, while encouraging you to become independent and confident in your skills.



Lower Learning Cost

Learning with us is budget-friendly. Our courses are affordable yet provide comprehensive and engaging content.



Expert Instructors

Learning with us means benefiting from top-tier expertise. Our expert instructors provide deep knowledge and engaging lessons.



Certification

Earn an industry-recognized certification upon completing the course, enhancing your resume.

C Programming

Introduction

- Why Programming
- History of C Language
- Hello World Program

Variables and Data Types

- Identifiers in C
- Variables and Data Types
- Constants

Console IO Operations

- printf and scanf
- Unformatted IO Functions

Operators and Expressions

- Expressions and Arithmetic Operators
- Relational and Logical Operators
- Bitwise Operators

Control Flow Statements

- If Statement
- Switch Statement
- Unconditional Branching using goto statement
- While Loop
- Do While Loop
- For Loop
- Break and Continue

Working with Functions

- Introduction and Writing Functions
- Scope of Variables, Storage Classes
- Pass by Value and reference
- Recursion

Working with Arrays

- Arrays Declaration and Initialization Properties
- Sample Programs using Arrays
- Arrays as Function Parameters
- 2-Dimensional Array

Pointers

- Introduction to Pointers
- Pointers as Function Parameter
- Pointer Arithmetic
- Pointers and Arrays
- Function Pointers
- Dynamic Memory Allocation using malloc
- calloc and comparison with malloc

String Handling

- Introduction to Strings
- Sample Program
- More Sample Programs
- Standard String Library Functions
- Array of String

Structures and Unions

- Declaring and Instantiating Structures
- Structure as Parameter and Pointer to Structure
- Enumerated Data Type
- Union
- Bit Fields

File Handling

- What is a Stream
- File Handling-Writing and Reading Characters
- Writing and Reading Structure in Text Format
- Writing and Reading in Binary Format

Pre-Processor Directives

- Understanding Pre-Processor directives
- Header Files and Project

Command Line Arguments and Variable Arguments

- Command Line Argument
- Variable Number of Arguments

Searching and Sorting

- Linear Search
- Iterative Binary Search
- Recursive Binary Search
- Bubble Sort
- Selection Sort
- Insertion Sort
- Quick Sort
- Merge Sort

Linked List

- Introduction to Linked List
- Single Linked List
- Double Linked List
- Circular Linked List

Stack

- Introduction to Data Structures
- Stack Using Arrays
- Stack Using Linked List
- Infix to Postfix Expressions

Queue

- Queue Using Arrays
- Queue Using Linked List

Trees

- Introduction to Trees
- Programming Tree – 1
- Programming Tree – 2

C++ Programming

Introduction

- History, Features
- Rules of C++ programming
- Structure of C++ program
- C++ Tokens
- Identifiers, Keywords, Constants, Operators, Special characters
- C++ Data types

Basic programming

- Console I/O Statements(cin, cout)
- Programs to perform various calculations
- Operators
- Programs to implement various operators

Control statements

- Conditional Control Statements
- If-else , switch-case
- Loops
- While, do while, for
- Implementing programs on conditional & loops
- break, continue, goto keywords

Arrays

- Definition, advantages
- Array types
- Single dimension
- Double dimension
- Declaration, accessing array data
- Implementation of array operations

Working with Functions

- Definition, advantages, types of functions, classification
- Implementing various kinds of functions
- Inline functions

Function Overloading

- What is function over loading
- Implementing overloading on various functions

Operator Overloading

- Definition, why it is required, advantages
- About operator keyword, rules of operator overloading
- Overloading various operators

Constructors & Destructors

- Definition, uses
- Types (Default Constructor, Parameter Constructor, Copy Constructor)
- Destructors

Friend Function & Friend classes

- Friend Function definition, usage of friend keyword
- Implementing of friend functions in various scenarios
- Friend Class definition, usage
- Implementing of friend class

Inheritance

- Definition, Advantages
- Types of Inheritances (Single, Hierarchical, Multilevel, Multiple Hybrid)
- Implementing various kinds of inheritances
- Implementing various constructors in inheritance
- Implementing function over riding
- Implementing various access levels in inheritance

Virtual Functions & Abstract Classes

- Definition, requirement
- How to implement
- Pure virtual function definition
- Implementing abstract classes

Templates

- Template Definition
- Generic Function
- Generic Class
- Template function Overloading

I/O Streams & Files

- Streams Hierarchy
- Input Streams & Output Streams
- What is File?
- Implementing various file operations on basic data types(write, read, append and modify)
- Implementing various file operations on object data types(write, read, append and modify)
- Random Access Files(seekp, seekg, tellp, tellg)

Formatted IO

- IO Flags
- IO Functions

Exception Handling

- Try, Catch and Throw

Thank You for Choosing **W3 Technologies**

Thank you for choosing to explore this course from W3 Technologies. We hope you find it insightful and valuable as you embark on your learning journey.

At W3 Technologies, we believe that learning is a continuous process, and we are excited to be part of your journey towards gaining new skills and knowledge.



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